Data Visualization Lab

Assignment 7

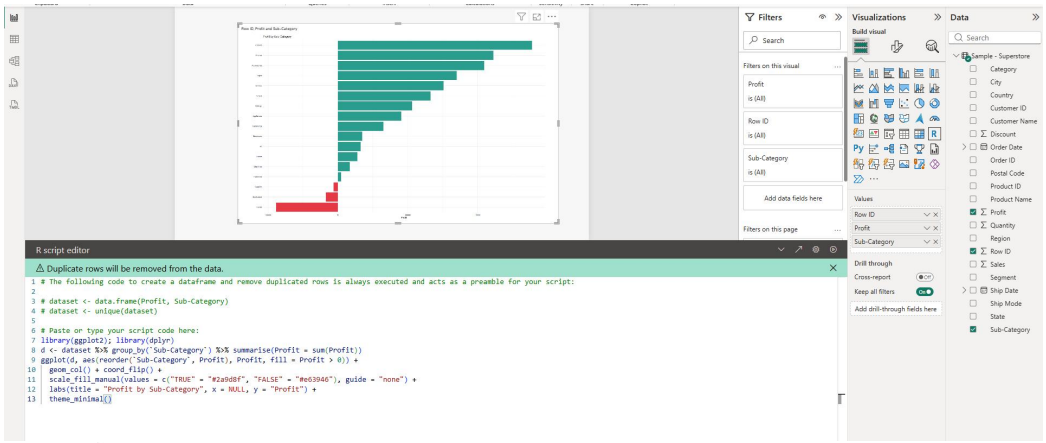
R Integration

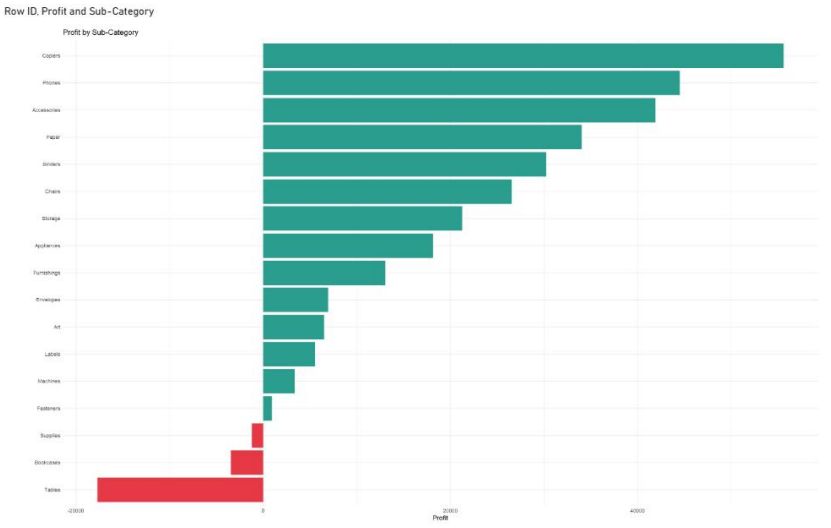
1. **R installation.** Installed R

2. **Import dataset.** I loaded the Sample Superstore CSV (9,994 rows, 21 columns) into Power BI

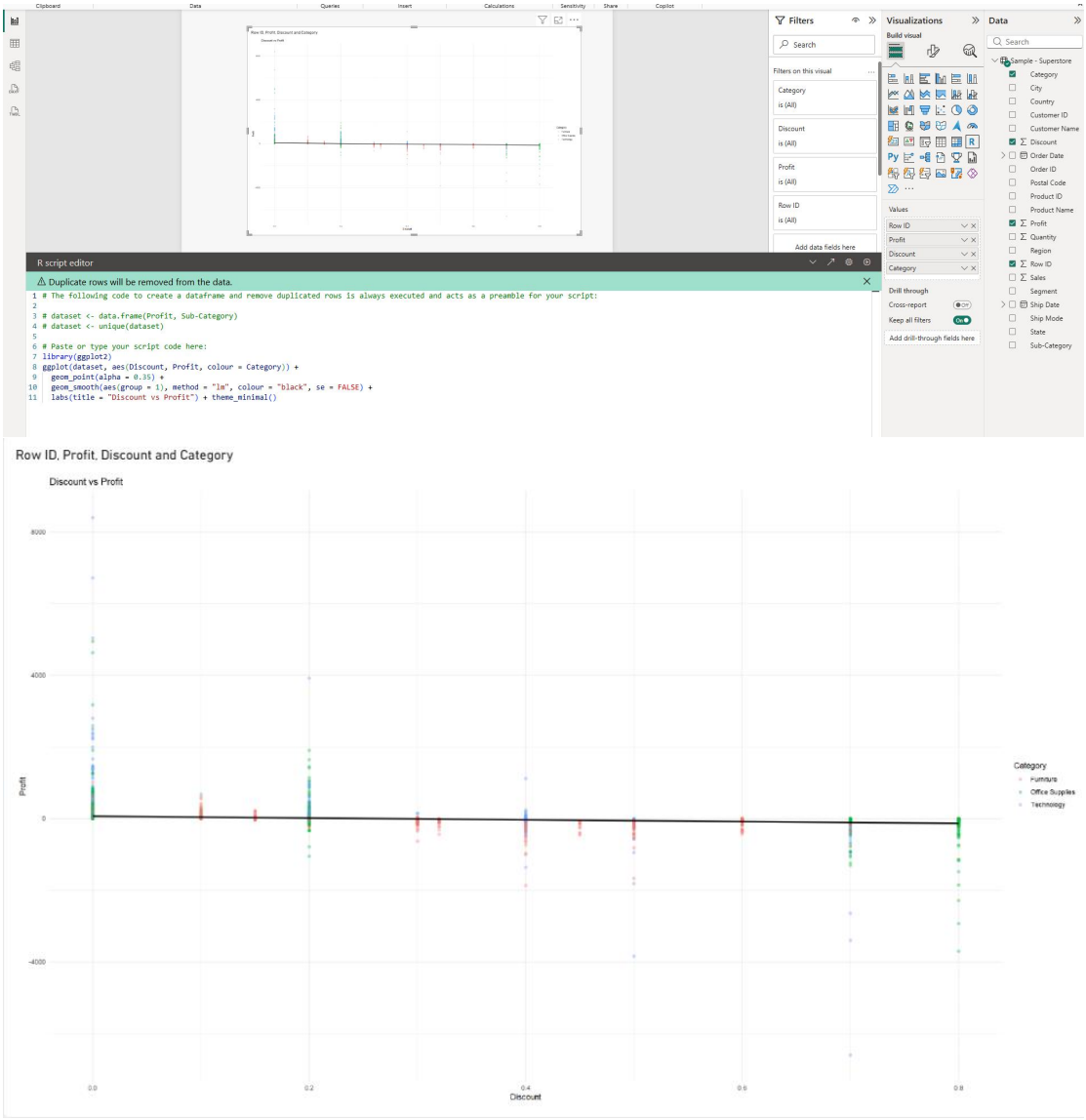
3. **Data cleaning.** Checked for missing values and duplicates and found none. Set the correct data types, such as Date for Order Date and numbers for Sales, Profit and Discount.

4. **R script.** Power BI using R script visuals and Power Query.

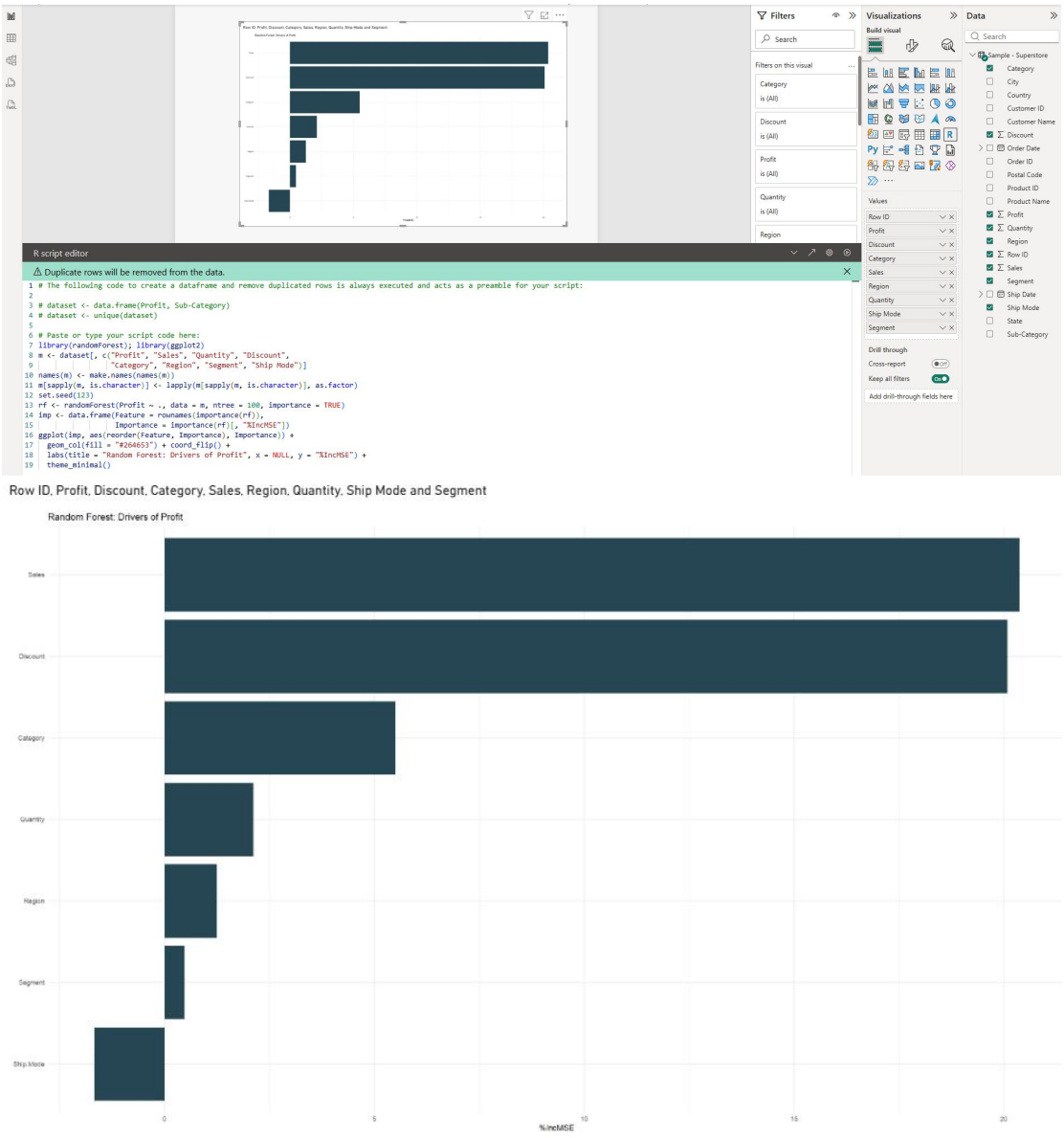




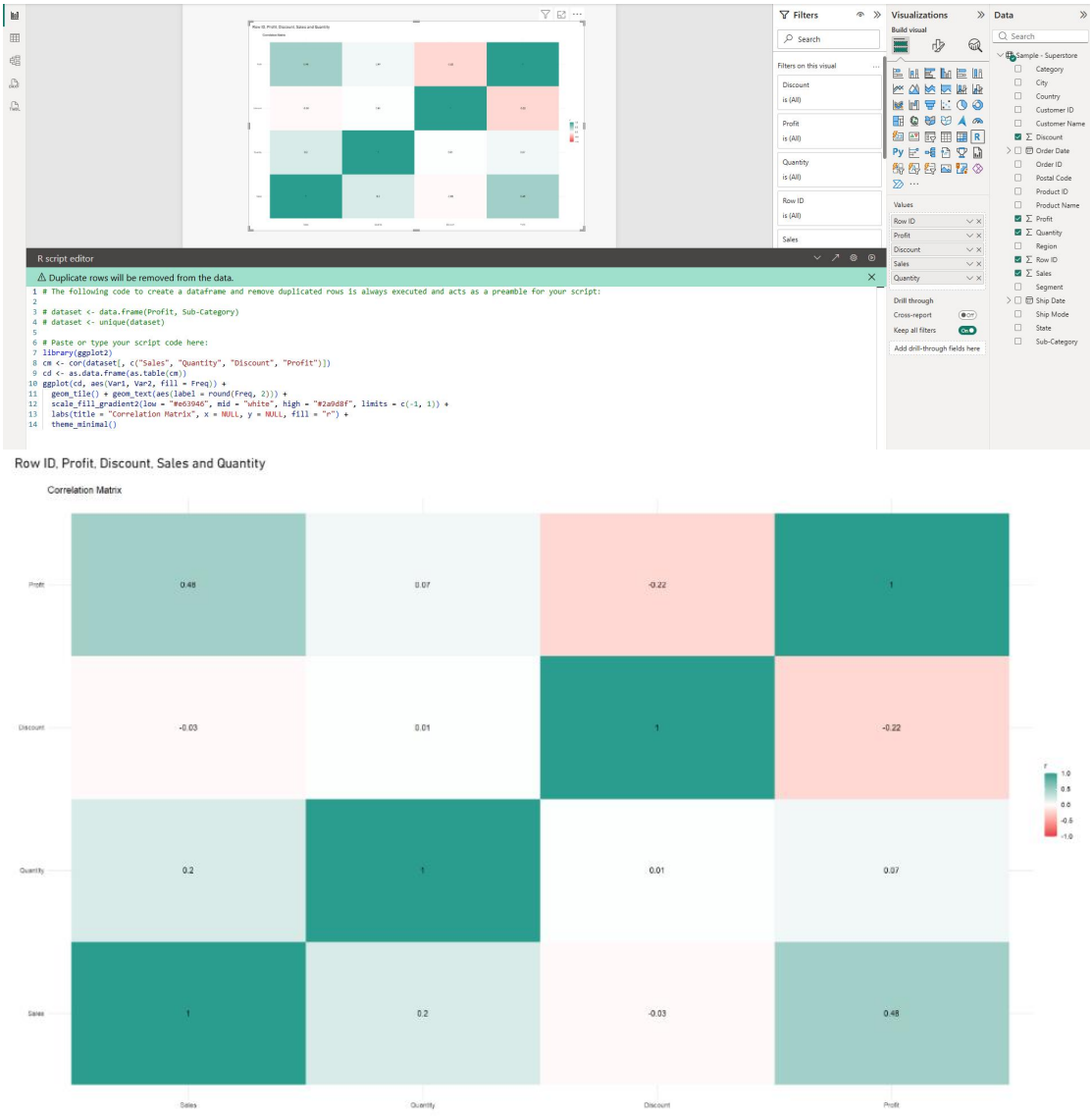
Profit by Sub-Category: Shows which product types make or lose money. Copiers, Phones and Accessories earn the most, while Tables, Bookcases and Supplies (in red) make a loss.



Discount vs Profit: Each dot is an order. The line slopes slightly down, which means bigger discounts tend to lower profit, and orders with 40% or more discount are often losses.



Random Forest: Drivers of Profit: Shows which factors the model relies on most to predict profit. Sales and Discount matter the most, Category matters somewhat, and Ship Mode and Segment matter very little.



Correlation Matrix: Shows how strongly the numbers move together, from -1 to 1. Profit rises with Sales (0.48) and falls as Discount rises (-0.22), while Quantity has almost no link to profit (0.07).

Total Sales
2.30M

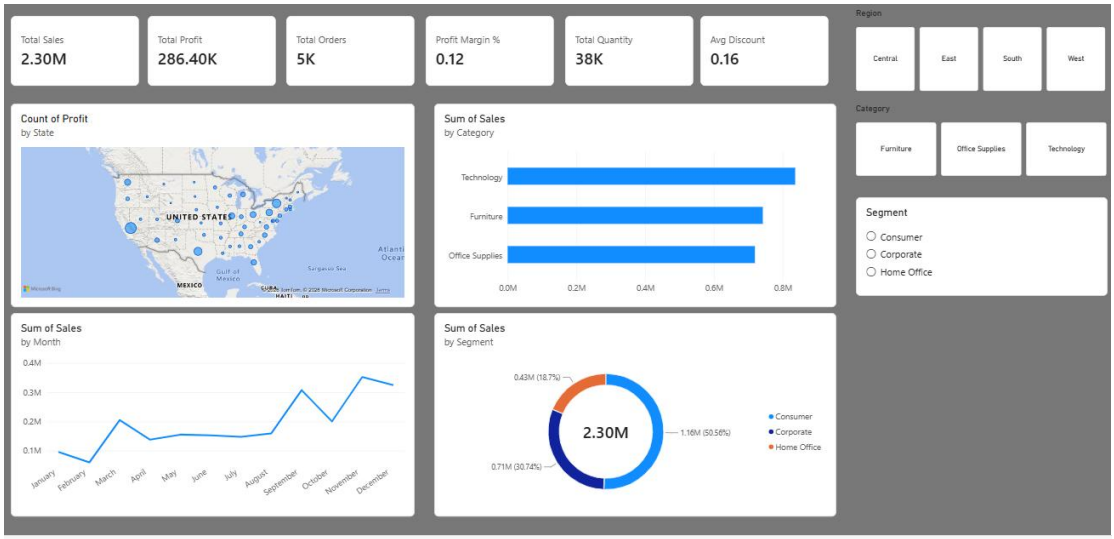
Total Profit
286.40K

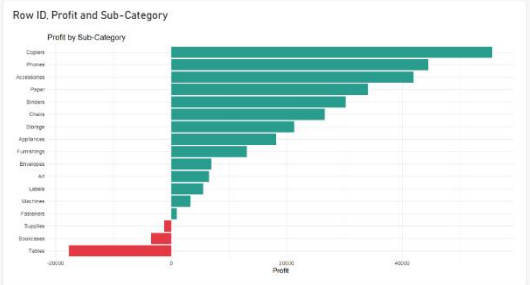
Profit Margin %
0.12

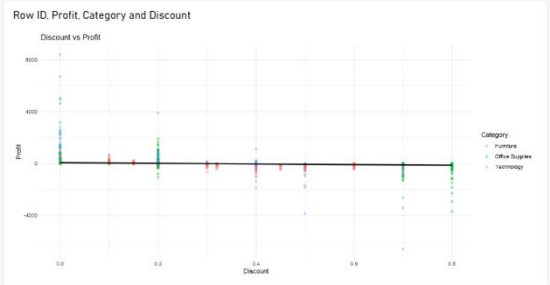
Total Orders
5K

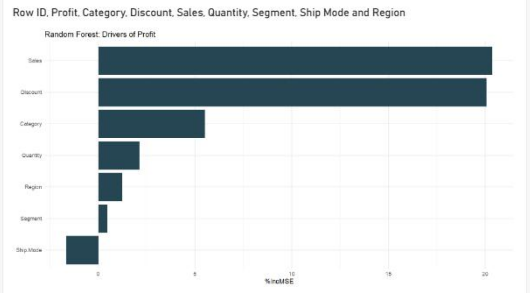
Total Quantity
38K

Avg Discount
0.16











Conclusion

The Random Forest model showed that sales and discounts drive profit the most. The business should limit heavy discounts, review pricing for Tables, Bookcases and Supplies, and plan stock for the busy months at the end of the year.